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Experiment-3: Decision Tree

Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

Python Program:

```
import pandas as pd

import matplotlib.pyplot as plt

from google.colab import files

from sklearn.preprocessing import LabelEncoder

from sklearn.tree import DecisionTreeClassifier, plot_tree

uploaded = files.upload()

filename = list(uploaded.keys())[0]

data = pd.read_csv(filename)

print("Dataset:")

print(data)

encoders = {}

for col in data.columns:

    le = LabelEncoder()

    data[col] = le.fit_transform(data[col])

    encoders[col] = le

X = data.iloc[:, 1:-1]
```

```
y = data.iloc[:, -1]

model = DecisionTreeClassifier(criterion='entropy', random_state=42)

model.fit(X, y)

plt.figure(figsize=(14,8))

plot_tree(

    model,

    feature_names=X.columns,

    class_names=encoders[data.columns[-1]].classes_,

    filled=True,

    rounded=True,

    fontsize=10

)

plt.title("Decision Tree using ID3 Algorithm")

plt.show()

new_sample = pd.DataFrame({

    "Outlook": ["Sunny"],

    "Temperature": ["Cool"],

    "Humidity": ["High"],

    "Wind": ["Weak"]

})

for col in new_sample.columns:

    new_sample[col] = encoders[col].transform(new_sample[col])

prediction = model.predict(new_sample)

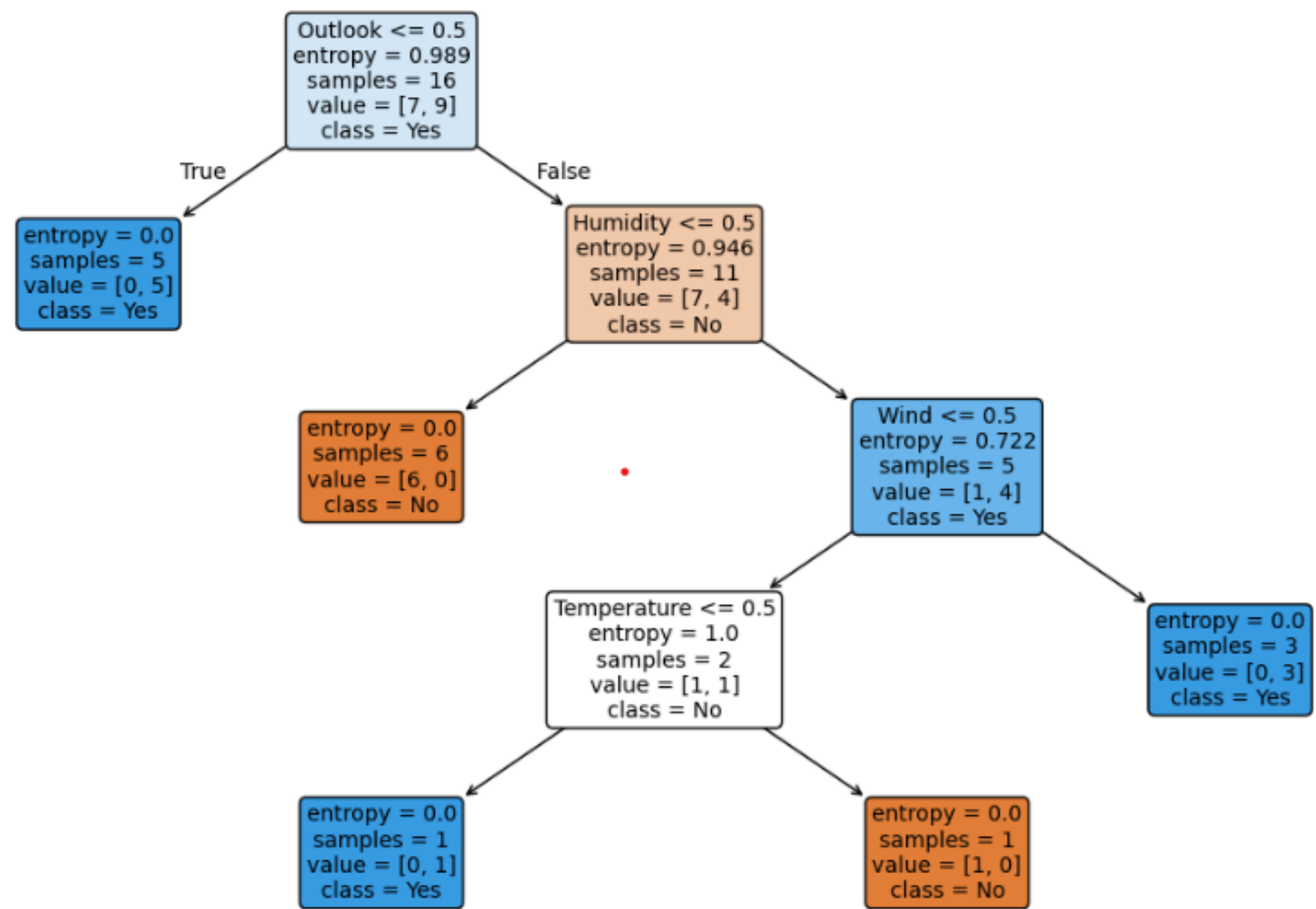
result = encoders[data.columns[-1]].inverse_transform(prediction)

print("\nPrediction for New Sample:")

print(result[0])
```

Output:

	ID	Outlook	Temperature	Humidity	Wind	Play
0	1	Sunny	Mild	Normal	Weak	Yes
1	2	Overcast	Cool	High	Strong	Yes
2	3	Rain	Hot	High	Weak	No
3	4	Sunny	Cool	Normal	Strong	Yes
4	5	Rain	Mild	Normal	Weak	Yes
5	6	Overcast	Hot	High	Weak	Yes
6	7	Sunny	Hot	Normal	Strong	No
7	8	Rain	Cool	High	Strong	No
8	9	Overcast	Mild	Normal	Weak	Yes
9	10	Sunny	Mild	High	Weak	No
10	11	Rain	Hot	Normal	Weak	Yes
11	12	Overcast	Cool	Normal	Strong	Yes
12	13	Sunny	Cool	High	Weak	No
13	14	Rain	Mild	High	Strong	No
14	15	Overcast	Hot	Normal	Strong	Yes
15	16	Sunny	Hot	High	Weak	No



Prediction for New Sample:
No